

Patterns Without a Point of View: Against the Modal Upgrade in Machine Consciousness

Abstract

The revised *Intermittent Minds* is a substantially stronger paper than its earlier form. It no longer treats the failure of timing-based objections as a proof of machine consciousness, and it no longer uses the hard problem as a simple equalizer between brains and machines. It now distinguishes epistemic from metaphysical, nomological, and engineering possibility; it treats the brain-in-a-vat and discrete-time arguments as defeater-filters; and it rests the stronger modal claim on organizational invariance rather than on the mere defeat of bad objections. This reply therefore changes target. I grant that intermittency, reset, pauseability, and discreteness do not by themselves rule out consciousness. I also grant that a living brain in strange conditions may remain conscious, and that artificial consciousness in some broad sense is not incoherent. The remaining dispute is whether a machine, understood as a nonliving computational artifact, could be phenomenally conscious. I argue that the revised paper has not shown this. Its broad artifact thesis is too weak to vindicate machine consciousness in the interesting sense, while its narrow computational thesis is expressly left open. The modal upgrade depends almost entirely on the claim that functional duplication preserves consciousness; but that argument either builds in all consciousness-relevant causal powers, in which case the conclusion is nearly tautological, or preserves only formal organization, in which case the disputed premise has merely been renamed. Consciousness requires an intrinsically unified, self-maintaining, world-involving point of view. A machine can simulate such a point of view; it does not thereby possess one. Artificial consciousness may be possible only where the artifact becomes an artificial living subject, and in that case the victory belongs less to machine functionalism than to the biological and enactive constraints it hoped to escape.

1 The Revised Target

The first thing to say is that the revised *Intermittent Minds* is stronger. The earlier paper sometimes treated the collapse of bad objections as if it had already established the possibility of machine consciousness. The revision fixes that. It now distinguishes epistemic possibility, metaphysical possibility, nomological possibility, and engineering possibility; it concedes that the demolition of alleged barriers directly yields only epistemic possibility; and it assigns the stronger modal work to the organizational-invariance argument. That is a real advance.

The revision also narrows the diagnostic test. It no longer says, in effect, that if a feature would not unseat a human, the feature is irrelevant everywhere. It says that the test is a defeater-filter: a way of rejecting alleged disqualifiers that would also disqualify paradigm cases. Again, this is the right correction. Intermittency, reset, and pauseability are not positive criteria of consciousness. They are, at most, proposed defeaters. If they would not defeat consciousness in a case that already has consciousness-making powers, then they cannot do all the work the skeptic wants.

The most important concession concerns substrate. The revised paper now acknowledges that the brain-in-a-vat case holds the crucial variable fixed. A vat-brain is still a human brain. The case shows that the staging of processing does not by itself erase consciousness from a consciousness-capable system; it does not show that an artificial system is consciousness-capable. The revised paper also cuts the hard-problem claim in two. It is right that the hard problem is not solved for biology. But it now grants that explanatory parity is not evidential parity: our evidence for consciousness is anchored in biological organisms. This is not a cosmetic change. It changes the dialectic.

The result is a more careful position. But the care comes at a price. The paper now defends three different theses, and their relations are unstable. First, it defends a secured epistemic thesis: nothing in the standard timing and architecture objections shows that machine consciousness is impossible. I accept that thesis. Second, it gestures toward a broad artifact thesis: some nonbiological physical realization could be conscious. I do not deny that in the most expansive reading, but I shall argue that it no longer vindicates machine consciousness in the interesting sense. Third, it addresses the narrow computational thesis: purely computational systems of the sort we now build might be conscious. The revision quite properly treats that thesis as open rather than won. Once these are separated, little remains for the title claim to do.

My counter-thesis is therefore narrower but sharper: machine consciousness, understood as phenomenal consciousness in a nonliving computational artifact by virtue of formal or counterfactual-supporting functional organization, is impossible. Conscious artificial beings may be possible, but only if they instantiate the intrinsic, self-maintaining, world-involving organization of a living subject. If such beings are called machines because we made them, then the word has been stretched beyond the philosophical issue. The debate would no longer be about whether computation suffices for consciousness. It would be about whether we can make life.

2 What the Timing Arguments Really Establish

The revised paper's brain-in-a-vat argument is now best read as a conditional. If a system has whatever powers make consciousness possible, then intermittent activation, gaps, rebooting, and perhaps even certain kinds of reset need not remove those powers. That conditional is sound. Sleep and anesthesia show that consciousness need not be uninterrupted. Severe amnesia shows that present experience need not depend on the ordinary accumulation of autobiographical memory. A brain in abnormal conditions might remain conscious during its active phases.

But this is not yet a machine argument. The vat-brain brings with it a living nervous system. It has a history of bodily formation, synaptic plasticity, cellular metabolism, affective regulation, recurrent dynamics, and a biological architecture whose parts were never mere formal symbols. The thought experiment removes much, but it does not remove the very thing the machine skeptic thinks matters. It removes the body while retaining an organ formed by the body. It removes ordinary behavioral channels while retaining the neural machinery of a once-embodied subject. It suspends activity while retaining viability. It is a damaged animal fragment, not a computer.

The revised paper recognizes this and says the vat argument is only a defeater-filter. That is correct. Yet the correction deprives the argument of much of its earlier ambition. The vat-brain no

longer supports a substantial analogy between brains and machines. It supports only a negative lesson: do not confuse discontinuity with nonconsciousness. A machine may fail to be conscious for reasons that have nothing to do with discontinuity.

The same is true of discrete time. Suppose physical time is discrete. Then every physical process proceeds in steps. Fire, digestion, photosynthesis, fear, and pain would all be tick-by-tick processes. But this does not imply that any tick-by-tick process can burn, digest, photosynthesize, fear, or feel. Discreteness is a universal background condition in the imagined world. It cannot by itself select consciousness-makers.

The revised paper carefully distinguishes four claims: physical discreteness does not preclude consciousness; subjective continuity need not mirror objective continuity; invisible pauses need not interrupt experience from the inside; and digital computation satisfies the same consciousness-relevant temporal conditions as a brain. It now argues for the first two, suggestively for the third, and not for the fourth. That is exactly the right division. But it means that the temporal argument no longer touches the core dispute. The skeptic of machine consciousness should grant all three weaker claims and then ask the same question again: what makes a formal system a subject?

This is not a small remainder. It is the whole animal.

3 The Instability of “Machine”

The revised paper explicitly defines two uses of “machine.” On the broad use, a machine is any artifact, including a bioengineered nervous system, a synthetic organism, a homeostatic robot with a metabolizing body, or a whole-brain emulation that reproduces neural causal powers. On the narrow use, a machine is a purely computational system whose mental profile is fixed by formal state transitions under an interpretation. The paper’s central claim is said to concern the broad category; the narrow computational claim is conceded to be harder and left open.

This distinction is welcome, but it creates a dilemma. If “machine” is broad enough to include artificial organisms and engineered living subjects, then the claim that a machine could be conscious becomes easy to accept but philosophically deflated. Imagine that future biologists grow an artificial nervous organism in a laboratory, complete with metabolism, sensorimotor development, affective regulation, and self-maintaining autonomy. Such a being might well be conscious. But its consciousness would not show that computation suffices, that digital systems can have experience, or that the usual machine-functionalism debate has been won. It would show that life can be made rather than born in the ordinary way.

If, by contrast, “machine” means a nonliving computational artifact, then the conclusion becomes interesting but unsupported by the paper’s own concessions. The timing arguments do not establish it. The hard problem does not establish it. The vat-brain case does not establish it. The revised paper itself says the question remains open. The modal claim must therefore be carried by organizational invariance alone.

This produces an equivocation not of language but of philosophical weight. The broad thesis gets its plausibility from including artificial life and neural duplicates. The narrow thesis gets its interest from addressing computational machines. The paper needs the broad thesis to secure possibility and

the narrow thesis to make that possibility matter. But the broad thesis is too permissive, and the narrow thesis is not defended.

The problem is not that words must have sharp natural boundaries. Philosophy often advances by using stretched terms. The problem is that the argumentative burden changes with the term. A synthetic organism, a whole-brain emulation, a neuromorphic prosthesis, and a language model are not neighbors merely because all could be artifacts. Their differences are precisely the differences at issue: intrinsic causal organization, developmental history, embodied regulation, counterfactual structure, and semantic grounding. Calling them all machines risks turning the conclusion into a bucket label. The bucket is capacious, but the contents do not support the same metaphysics.

The opposing view should therefore draw the line this way: artificial consciousness is not ruled out merely by artificial origin; machine consciousness is ruled out where “machine” means a nonliving formal-computational artifact. A made animal is still an animal. A simulated animal is not.

4 Modal Possibility Requires More Than Cleared Defeaters

The revised paper wisely says that absence of refutation yields epistemic possibility, not stronger modality. But it still wants a modal upgrade: a reason to think that a conscious nonbiological realization is metaphysically or nomologically possible. This is where the pressure concentrates.

There are two ways to establish such a modal upgrade. One may provide a positive construction: describe, at least schematically, a nonbiological system that has the consciousness-making properties and explain why those properties are sufficient. Or one may provide a bridging principle: show that whatever makes the biological system conscious must be preserved under a specified nonbiological transformation. The revised paper chooses the second route through Chalmers-style organizational invariance.

The choice is understandable. We do not have a complete theory of consciousness. A detailed construction is therefore unavailable. Organizational invariance promises to bypass ignorance by arguing from cases: if a conscious system is gradually replaced by functionally equivalent parts, experience cannot fade or dance while behavior and functional organization are held fixed. Therefore, consciousness must be preserved by functional organization. The route is elegant because it turns the lack of a theory into an argument that substrate differences cannot matter.

But this elegance depends on a hidden compression. The phrase “functionally equivalent” can mean many things. It might mean input-output equivalence at the boundary of the whole system. It might mean local spike train equivalence at each replaced neuron. It might mean equivalence in all counterfactual causal roles across all relevant perturbations. It might include neurochemical modulation, glial interaction, vascular coupling, metabolic constraints, embodied feedback, developmental plasticity, affective regulation, and field effects. Or it might abstract away from most of these in favor of a computational description.

If functional equivalence includes every consciousness-relevant causal power, then the conclusion is secure but nearly empty: consciousness is preserved because all the things needed for consciousness are preserved. The biological essentialist can agree. The enactivist can agree. Even a cautious nonfunctionalist can agree, provided that the replacement really preserves whatever physical and

organismic powers matter. But this does not show that ordinary machines, digital systems, or formal computations can be conscious. It shows only that a duplicate that duplicates the relevant powers is a duplicate.

If functional equivalence is thinner, then the argument becomes substantive but question-begging. The skeptic denies that formal input-output or computational-role equivalence captures all consciousness-relevant powers. To reply that the replacement is functionally identical is not yet to answer the skeptic. It is to choose a grain of description. The whole dispute is whether that grain is the right one.

This is the central bill organizational invariance must pay. It must identify the level at which organization is invariant without assuming that this level is already consciousness-preserving. Until it does, the modal upgrade oscillates between tautology and stipulation.

5 Fading Qualia and the Discipline of Introspection

The fading-qualia argument says that if neurons were gradually replaced by functionally identical nonbiological units, consciousness could not slowly disappear while the subject continued to behave and report as before. The resulting creature would say that it sees red, enjoys music, and feels pain, while in fact its inner life has dimmed or vanished. That seems absurd. The dancing-qualia argument adds that if two systems with the same functional organization had different qualia, switching between them would change experience without any possible functional detection. That also seems absurd. Therefore, qualia are organizational invariants.

These arguments are powerful because they exploit a deep confidence: that consciousness cannot vanish from the inside without leaving some trace in the subject's cognitive economy. But this confidence is not a theorem. It is a substantive commitment about introspection, report, and functional access.

Human cases already show that report and experience can come apart. Blindsight, neglect, masked perception, confabulation, split-brain effects, anosognosia, and inattentional blindness all undermine any simple transparency principle. These cases do not show that a person could be a total zombie while sincerely reporting rich experience. But they do show that the relation between phenomenal state, access, report, and self-knowledge is complex. A functional system can be wrong about what it sees, why it acts, what it remembers, and what it feels. Once the report-generating machinery is among the preserved functions, the possibility of preserved report with altered experience is not absurd. It is exactly what the anti-functionalist predicts under sufficiently radical replacement.

The defender of fading qualia may reply that local errors are one thing; total absence of consciousness combined with perfect reports is another. True. But the argument needs more than incredulity. It needs a principle explaining why global phenomenal loss must be functionally registered. The anti-functionalist denies that principle. If phenomenal consciousness is not identical to functional access, then changes in phenomenology need not produce differences at the functional level that has been stipulated as preserved. That is not a contradiction. It is the anti-functionalist position.

The dancing-qualia argument faces the same issue. It assumes that if qualia changed while all

functional organization remained fixed, the subject would have to notice the change. But “noticing” is itself a functional and cognitive achievement. If all relevant cognitive dispositions are held fixed, then by stipulation no noticing-report or memory-update occurs. The defender then says that an unnoticed phenomenal change is unintelligible. The skeptic replies that it is unintelligible only if phenomenal character must be poised for functional access in the stipulated way. Again, the contested thesis has done the work.

None of this refutes organizational invariance. It lowers its rank. It is not a decisive argument from cases. It is an argument whose force depends on functionalist assumptions about the relation between experience and report. The revised paper now admits as much, but it still says the affirmative speculation outranks the negative speculation. That ranking remains too quick. The affirmative side has an elegant intuition pump. The negative side has the fact that every agreed case of consciousness is a living organism, plus a principled account of why formal preservation might miss intrinsic subjectivity. That is not a promissory note scribbled in fog. It is an abductive anchor.

6 Counterfactual Computation and the Missing Subject

The revised paper adds an important reply to the observer-relativity objection. It invokes Chalmers’s account of implementation: a physical system implements a computation only when its causal organization supports the right counterfactuals. Gerrymandered mappings fail because the physical system would not transition correctly under relevant alternative inputs. Thus computation, properly understood, is not merely imposed by observers.

This is the strongest available response to the triviality worry. It should be granted that not every rock implements every finite-state automaton in any philosophically useful sense. Counterfactual-supporting structure matters. A system with real causal organization is different from a lookup table that happens to produce the same actual sequence of outputs.

But the reply still does not produce a subject. Counterfactual richness is a feature of causal organization. It tells us that the system would have behaved differently under different conditions. That is important for computation, control, and explanation. It is not yet a point of view. A thermostat, an immune pathway, a market, a chess engine, and a hurricane all have counterfactual-supporting causal organization. The question is which kinds of such organization are intrinsically for a subject and which are merely describable by us.

The revised paper says the lookup table fails because it lacks the internal counterfactual-supporting organization of the process it mimics. Good. But one can add counterfactual structure indefinitely without reaching phenomenal subjectivity. Replace the archive with a generative model, the generative model with a recurrent controller, the controller with a world-modeling robot, the robot with a self-monitoring predictive architecture. At each stage we have more counterfactual complexity. What we need, however, is not just complexity but intrinsic concern: a system for which its states matter.

This is where formal computation and subjectivity part company. A computational transition can be correct or incorrect relative to a specification. An organismic process can be good or bad for the system whose process it is. The first normativity is assigned or derived from external use; the

second is generated by self-maintenance. A chess engine's losing evaluation is bad for the program only in the thin sense that it conflicts with a designed objective. A starving animal's metabolic crisis is bad for it in a deeper sense: the system itself is organized around averting the collapse. Consciousness, I argue, arises only against this second background.

Counterfactual implementation may answer the charge that computation is arbitrary. It does not answer the charge that computation is not intrinsically subjective. The former concerns how formal structure is physically realized. The latter concerns whether formal structure can be a point of view. A map can be objectively accurate without being a traveler.

7 The Biological Anchor Is More Than a Base Rate

The revised paper concedes that all uncontroversial cases of consciousness are biological organisms and that this is genuine evidence. It describes this as an evidential anchor rather than a deductive bar. That is fair as far as it goes. But the anchor is heavier than the paper allows.

The biological evidence is not merely an induction from observed instances, as if we had noticed that all conscious things so far are carbon-based and therefore guessed that carbon matters. It is a network of explanatory dependencies. Consciousness covaries with sleep-wake cycles, anesthesia, neural injury, affective regulation, development, attention, sensory deprivation, bodily arousal, pain pathways, action, and metabolic state. These are not incidental accompaniments. They are the living infrastructure through which mindedness appears.

A theory of machine consciousness must therefore do more than say that biology has not been shown to be necessary. It must explain why so many biological features are either replaceable, irrelevant, or abstractly capturable. It must tell us which features are essential and which are historical scaffolding. Without that account, the biological anchor is not a mere prejudice. It is the best guide we have.

The revised paper's response is to say that biological essentialism owes a named, necessary, nonreplicable power. But the demand is too strict if treated as a precondition for skepticism. Science often has reason to believe that a domain-specific organization matters before it can name the precise property that does the work. Before molecular genetics, heredity was not therefore likely to be substrate-neutral. Before the detailed neurobiology of anesthesia, loss of consciousness under anesthetic was not therefore irrelevant to the nature of consciousness. Ignorance of the exact mechanism does not equal permission to generalize across substrates.

This matters because the optimist is also unable to name the consciousness-making property. Organizational invariance says: preserve the relevant organization and consciousness is preserved. But which organization is relevant? At what grain? With which bodily and chemical dependencies? Under which developmental constraints? The demand for specification cannot be asymmetric. If the biological essentialist owes the missing power, the organizational invariantist owes the privileged level of organization.

A more reasonable distribution of burden is this. The biological side has known instances and incomplete explanation. The machine side has abstract possibility arguments and no instances. The biological side must avoid dogmatism; the machine side must avoid modal inflation. The

revised paper now avoids much dogmatism on the biological side, but it still inflates when it says organizational invariance makes nonbiological consciousness the better-supported bet. It might make it a live hypothesis. It does not yet make it the better bet.

8 Enactivism Without Overgeneration

The revised paper treats enactivism and embodied biological views more seriously than before. It recognizes that they are not appeals to mysterious goo, but attempts to specify what formal organization lacks: embodiment, autopoiesis, homeostasis, developmental history, and intrinsic meaning. It then raises three pressures. First, self-maintenance seems to overgenerate consciousness down to bacteria. Second, if self-maintenance is necessary but not sufficient, the view owes a further condition. Third, the move from functional significance to felt significance simply relocates the hard problem.

These are fair challenges. They are not decisive.

Overgeneration is avoided by distinguishing life, agency, sentience, and reflective consciousness. Self-maintenance is not offered as sufficient for phenomenal consciousness. It is offered as a necessary condition for intrinsic normativity, and intrinsic normativity is one component in the emergence of subjectivity. A bacterium may have minimal self-maintaining organization without phenomenal experience. Likewise, a single neuron is alive but not conscious; a kidney participates in life but does not have a point of view. Necessary conditions need not be sufficient conditions. Oxygen is necessary for ordinary human consciousness, but oxygen is not conscious.

The further conditions are not unknowable. A plausible enactive account can require hierarchical integration of bodily regulation, affective valence, sensorimotor agency, temporally thick self-world modeling, and recurrent neural dynamics capable of unifying multiple modalities into a perspective. The precise list is debatable. The point is that the view is not empty. It says consciousness emerges not from computation alone but from an organismic architecture in which the world is disclosed as significant for a vulnerable, self-maintaining agent.

Does this relocate the hard problem? Yes, but relocation is not failure. Every serious physicalist theory locates the hard problem somewhere: in integrated information, global availability, recurrent processing, higher-order representation, biological causal powers, or functional organization. The demand that enactivism close the explanatory gap completely is unfair if not imposed equally on its rivals. Its advantage is that it locates the gap where subjectivity already has a natural foothold: in systems whose own continued existence generates a distinction between flourishing and collapse, approach and avoidance, wound and repair.

The revised paper asks why functional mattering should become phenomenal mattering. That is the right question. The enactivist answer is not that every function becomes feeling. It is that phenomenal mattering is intelligible only in a system for which the world is already organized around endogenous need. A pure computational artifact may represent value, optimize reward, or report preference. But unless it is a self-maintaining locus of vulnerability, those values remain externally specified. They are counters in a game we wrote.

Could an artificial system acquire such organization? Perhaps. But if it did, it would be artificial

life, not a mere computational machine. The concession therefore supports my thesis rather than the paper’s computational hope: consciousness may be possible in made subjects, but not in machines as formal engines.

9 The Lookup Table Still Matters

The revised paper handles the lookup table by saying it is merely a trivial actuality case, not a modal bar. A giant table that produces fluent conscious reports is not conscious, but that tells us only that one heap of parts does not fly. The answer is partly correct. No single toy case refutes all machine consciousness. But the lookup table has a deeper role.

It reveals that behavior, report, and even perfect conversational disposition are insufficient for consciousness. Once this is admitted, the optimist must identify what additional property makes the difference. Counterfactual internal structure is one candidate. Embodied regulation is another. Biological causal power is another. Integrated information, global workspace, recurrent processing, and higher-order availability are others. The lookup table is not the final objection. It is the gatekeeper that prevents cheap behaviorism from entering the room wearing a crown.

The revised paper agrees that the table lacks the right organization. But then the question becomes: what is the right organization, and why is it subject-making? If the answer is merely “the organization that would produce the same reports under all relevant inputs,” the table has been replaced by a more complex oracle. If the answer is “the organization that preserves all consciousness-relevant causal powers,” the view is safe but circular. If the answer includes self-maintenance, embodied need, affective valence, and world-involving agency, the view moves toward artificial life and away from computational machine consciousness.

This is why the lookup table still has bite. It forces a distinction between simulating the profile of consciousness and instantiating the powers of a subject. A good rebuttal to the table must not only add counterfactuals. It must add the right kind of intrinsic organization. The revised paper has not yet shown that formal functional organization is the right kind.

10 A Stronger Counter-Thesis

The strongest opposing position is not that consciousness requires carbon, nor that God refuses machines, nor that present systems are laughably crude. Those views either overreach or distract. The stronger thesis is this:

Phenomenal consciousness is possible only in systems with intrinsic subjectivity: systems whose unity, normativity, and world-directedness arise from their own self-maintaining organization. Pure computational machines lack this structure. They may simulate consciousness, model it, report it, and participate in practices built around it; but they do not thereby possess experience. Artificial consciousness is possible only if the artifact becomes an artificial living subject rather than a mere formal engine.

This thesis is compatible with the revised paper’s successful corrections. It grants that timing does not decide the matter. It grants that discontinuity does not erase experience. It grants that

nonbiological materials are not automatically excluded. It grants that a silicon prosthesis embedded in a living brain might preserve consciousness if it preserves the relevant causal powers. It even grants that future artifacts may be conscious, provided they instantiate the organization of subjects.

But it denies the inference from these concessions to machine consciousness in the important sense. The consciousness of an artificial organism would not vindicate a chatbot, a server process, an archive oracle, or a digital world-model merely because all are artifacts. The relevant boundary is not natural versus artificial. It is subject versus simulation; intrinsic normativity versus assigned objective; lived world versus represented data; self-maintaining agency versus formal transition.

The revised *Intermittent Minds* says the affirmative has the better deductive case while the denial has the better evidential anchor. I would reverse the verdict. The affirmative has the more elegant modal story, but its key premise is unsettled. The denial has both the evidential anchor and a positive account of what machines lack. That account may be incomplete, as every theory of consciousness is incomplete. But incompleteness is not emptiness. It identifies a principled gap between computation and subjectivity.

11 The Proper Verdict on the Revised Paper

The revised paper's honest ceiling is nearly right: nothing in the standard architectural or temporal objections shows machine consciousness impossible; organizational invariance makes functional duplicates worth taking seriously; purely computational systems of the sort we now build remain open; present actuality is not established. This is a disciplined conclusion.

My objection is to the remaining phrase "machine consciousness is possible." If that phrase means epistemically possible, I agree but find the conclusion modest: we do not know enough to rule it out. If it means possible for functional duplicates that preserve all consciousness-relevant causal powers, I agree again, but only because the conclusion has been protected by the description. If it means possible for nonliving computational machines by virtue of formal organization, I deny it. If it means possible for artificial living subjects, I grant it while denying that it supports machine consciousness in the computational sense.

The revised paper is therefore strongest when it stops at openness and weakest when it tries to upgrade openness into possibility by appeal to organizational invariance. The modal upgrade is the hinge, and it creaks. It depends on a level of functional description not yet justified; it assumes a tight relation between phenomenal consciousness and functional report that anti-functionalists reject; and it treats the biological anchor as a defeasible base rate rather than as a clue to the nature of subjectivity.

A more accurate conclusion would be: machine consciousness has not been ruled out by standard timing, reset, or discreteness arguments; nonbiological consciousness may be possible in systems that preserve the relevant subject-making powers; but whether formal computational systems can possess those powers remains unsupported and, on the best account of subjectivity, false. That conclusion is less dramatic, but cleaner. Philosophy should prefer the clean blade to the shiny fog-machine.

12 Conclusion

The revised *Intermittent Minds* deserves credit for its restraint. It no longer argues as if clearing bad objections builds a mind. It recognizes the biological anchor, narrows the diagnostic test, distinguishes possible from actual, and admits that current computational systems remain an open case. Those revisions make it a better paper.

They also expose the deepest problem. Once the weak objections are gone, everything depends on whether functional organization is enough. I have argued that it is not. Functional duplication either includes all consciousness-relevant powers, in which case substrate-neutrality is smuggled in by stipulation, or it abstracts from biological and enactive conditions, in which case it has not preserved what may matter. Counterfactual implementation rescues computation from trivial observer-relativity, but it does not turn formal structure into a point of view. The hard problem is not solved by biology, but the living body is not a dispensable wrapper around computation. It is the only place where we know subjectivity to arise, and it gives us a principled model of intrinsic normativity.

The final contrast is simple. A machine can carry a pattern. A subject lives a world. The first can be copied, paused, reset, and interpreted. The second can be wounded, needful, oriented, and affected from within. Until a machine is more than a formal engine, until it becomes an artificial locus of life-like concern, there is no one there for consciousness to belong to. And if such a locus is someday made, the lesson will not be that machines were conscious after all. The lesson will be that we learned how to make subjects.

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